

# Comparative Evaluation of Classical, Deep Learning, and Quantum-Enhanced Architectures for Lung Cancer Classification from CT Images

*A controlled four-paradigm benchmark — CNN, SVM, Vision Transformer, and Hybrid Quantum Vision Transformer  
— on the IQ-OTH/NCCD CT dataset*

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# Lung Cancer: A Global Diagnostic Challenge

**~1.8M**

Deaths worldwide per year — the leading cause of cancer mortality

**18%**

Of all cancer-related deaths globally are attributable to lung cancer

**<20%**

Five-year survival rate — driven by late-stage detection

**1mm**

Axial CT slice thickness enabling sub-centimeter nodule detection

## Why Computer-Aided Diagnosis?

- CT remains the established modality for lung cancer screening, offering high-resolution detection of sub-centimeter nodules
- Accurate interpretation requires specialised radiological expertise, is time-intensive, and is subject to inter-examiner variability
- These constraints have driven sustained interest in machine learning-based computer-aided diagnosis (CAD) systems

# The Research Gap

| Era                  | Method                                              | Advance                                 | Constraint                                  |
|----------------------|-----------------------------------------------------|-----------------------------------------|---------------------------------------------|
| Classical (pre-2015) | SVM + handcrafted features (GLCM, Gabor, radiomics) | Strong margin-based generalisation      | Descriptor design requires domain expertise |
| Deep Learning        | Convolutional Neural Network (CNN)                  | Learns hierarchical features end-to-end | Local receptive fields miss global context  |
| Transformer          | Vision Transformer (ViT)                            | Captures long-range dependencies        | Weak spatial bias on small datasets         |
| Quantum-Enhanced     | Hybrid Quantum ViT (QViT)                           | Novel representational capacity         | Constrained by current NISQ hardware        |

No prior study has evaluated CNN, SVM, ViT, and QViT together under one controlled, balanced protocol.

RQ1: Which architecture achieves the highest performance under a controlled protocol, and can QViT compete with established baselines?

RQ2: How do the four paradigms differ in feature-learning and interpretability, and what trade-offs do these present for clinical deployment?

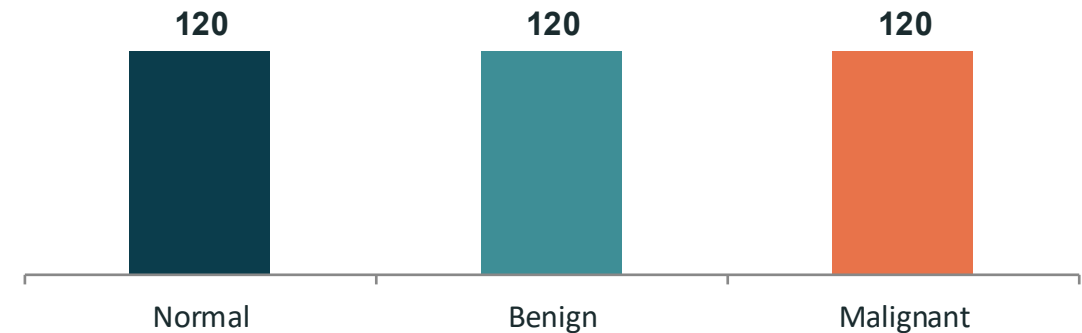
# Dataset & Class Balancing

## IQ-OTH/NCCD Dataset

- 1,190 thoracic CT images from 110 patients
- Iraqi Oncology Teaching Hospital & National Center for Cancer Diseases
- Siemens SOMATOM scanners, 1mm axial slice thickness
- Three diagnostic categories
- Substantial class imbalance: Malignant and Normal slices considerably outnumber Benign cases

## Class Balancing (Stratified Downsampling)

Balanced Dataset — 360 CT Images (120/class)



# A Unified Experimental Protocol

*Every design choice below is held identical across all four models — isolating architecture as the sole source of performance differences.*

01

## Preprocessing

224×224 resolution; ImageNet normalization ; horizontal flip — the sole anatomy-preserving augmentation

02

## Data Split

Stratified 70/30 split, seed=42 →  
252 training images (84/class), 108  
held-out test images (36/class)

03

## Cross-Validation

4-fold stratified CV within training  
set: 189 train / 63 validation images  
per fold

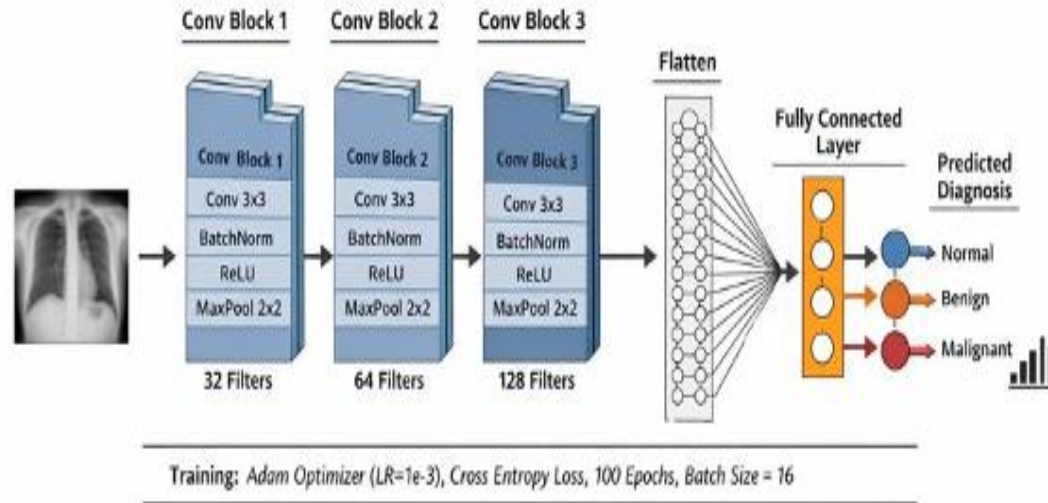
04

## Evaluation

Macro-averaged precision, recall,  
F1, and one-vs-rest ROC-AUC —  
equal weight per class regardless of  
support

# Model Detail: CNN

## CNN — Convolutional Neural Network



**Method:** Three-block Convolutional Neural Network.

**Role:** Learns hierarchical spatial features directly from CT images for lung cancer classification.

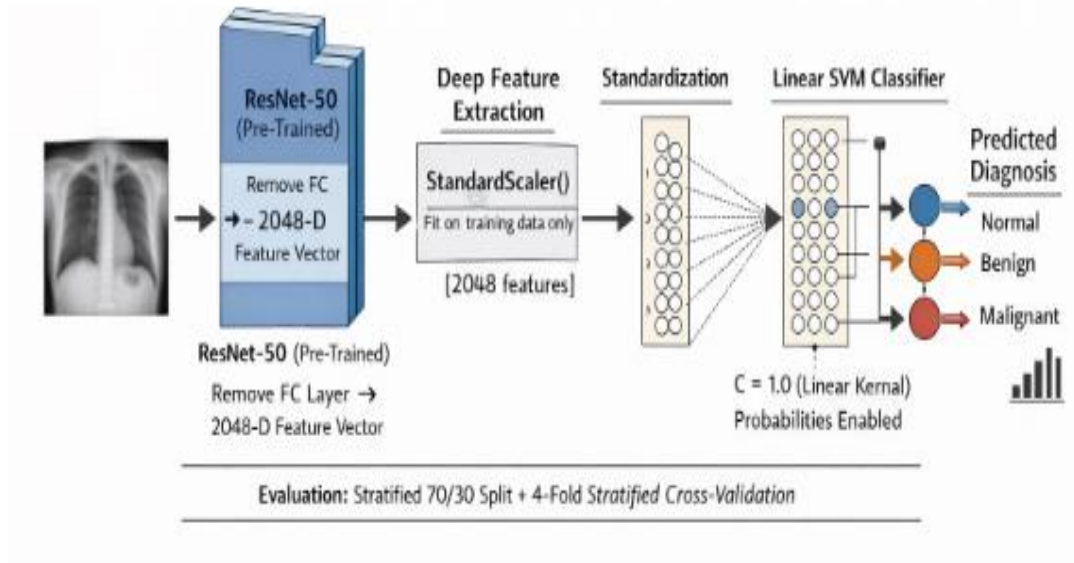
**Strength:** Effectively captures local texture and nodule characteristics, achieving the highest overall classification performance.

### Design Rationale

Convolutional weight-sharing and pooling encode the 5–30 mm spatial scale of pulmonary nodules, giving the CNN an inductive bias suited to a 252-image training set. BatchNorm after each convolution reduces internal covariate shift and provides implicit regularisation, limiting overfitting in the compact 256-unit fully connected layer.

# Model Detail: SVM

## SVM — ResNet-50 + Linear SVC



**Method:** ResNet-50 deep feature extractor with a Linear SVM classifier

**Role:** Uses pretrained 2048-dimensional deep features to classify CT images using an optimal decision boundary.

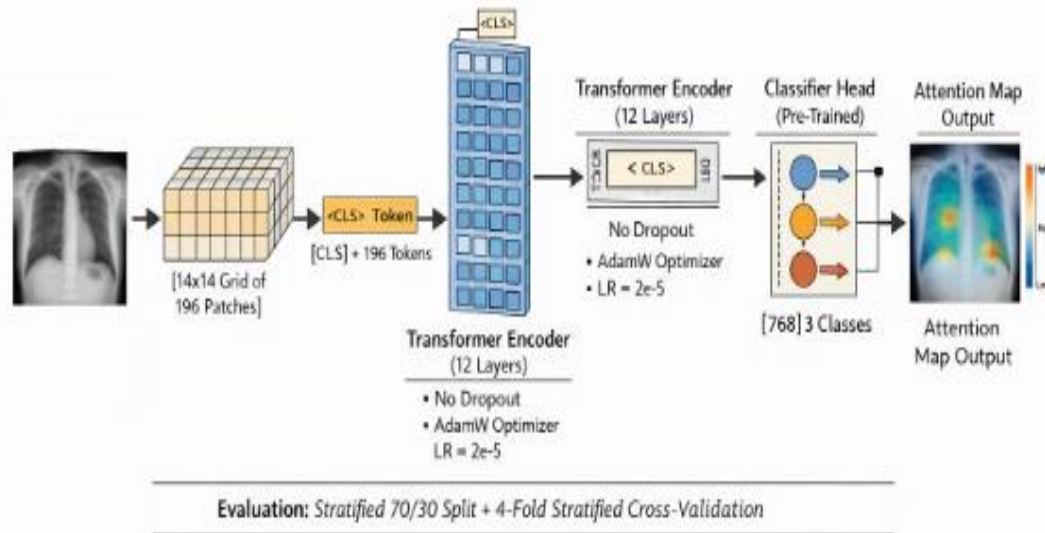
**Strength:** Computationally efficient and highly effective for small medical imaging datasets while achieving competitive performance.

### Design Rationale

Training a deep network end-to-end on only 252 images risks severe overfitting. The SVM sidesteps this by pairing a frozen, semantically rich ResNet-50 representation with a simple linear decision boundary — the deep features alone lift accuracy from 89.9% to 92.59% over handcrafted descriptors.

# Model Detail: ViT

## ViT — Vision Transformer



**Input:** 224×224 CT slices split into 196 non-overlapping 16×16 patches, linearly embedded with a prepended CLS token

**Mechanism:** 12-layer multi-head self-attention computes pairwise relationships across all 197 tokens from layer one — global context from the start, not built up gradually

**Strength:** Directly compares left-vs-right lung symmetry and lung-to-mediastinum relationships from the first layer, rather than building this up gradually through depth like a CNN.

### Design Rationale

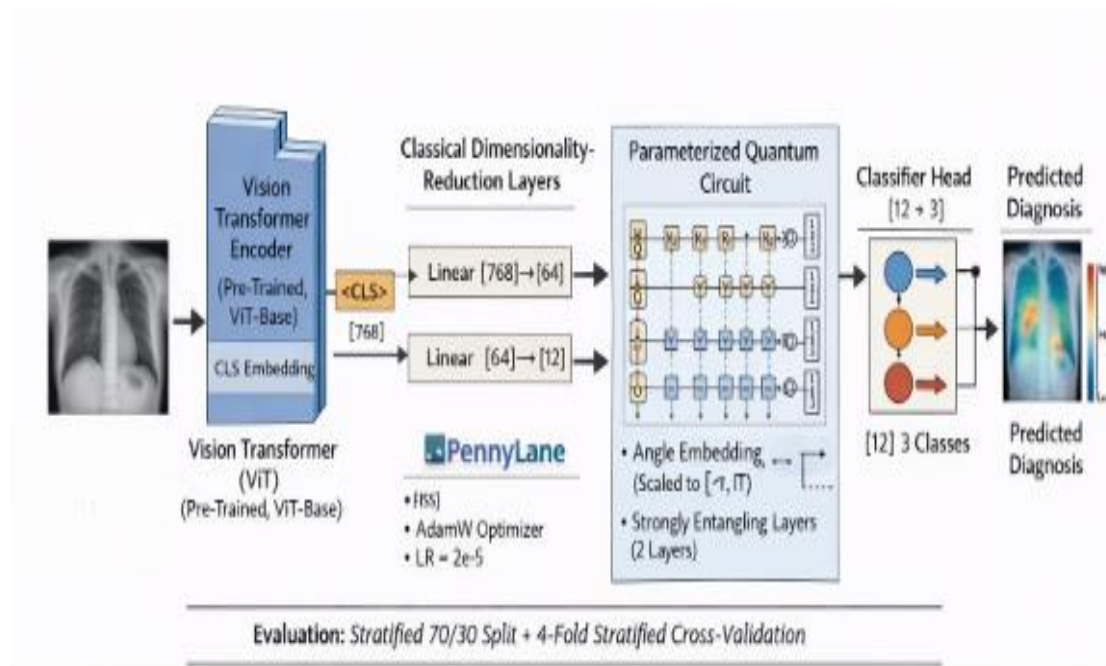
#### Pretrained ViT: google/vit-base-patch16-224-in21k

Self-attention has no built-in spatial bias — a real drawback when training from scratch on just 252 images.

ViT sidesteps this by fine-tuning from a large ImageNet-21k prior (21,843 categories) at a cautious  $2 \times 10^{-5}$ , letting pretrained attention patterns to do the work.



# Model Detail: QViT



## QViT — Hybrid Quantum Vision Transformer

**Input:** The pretrained ViT's 768-dimensional CLS embedding is compressed through a two-stage classical bottleneck ( $768 \rightarrow 64 \rightarrow 12$ ) before entering a 12-qubit circuit.

**Mechanism:** StronglyEntanglingLayers applies rotations across all 12 qubits (72 trainable quantum parameters) with cyclic entanglement; Pauli-Z expectation values feed the final 3-class classifier.

**Strength:** Despite a 98%+ compression before quantum encoding, QViT still reaches a perfect Malignant AUC (1.00) — proof the quantum circuit's ranking signal is sound even where NISQ-era hardware limits its capacity.

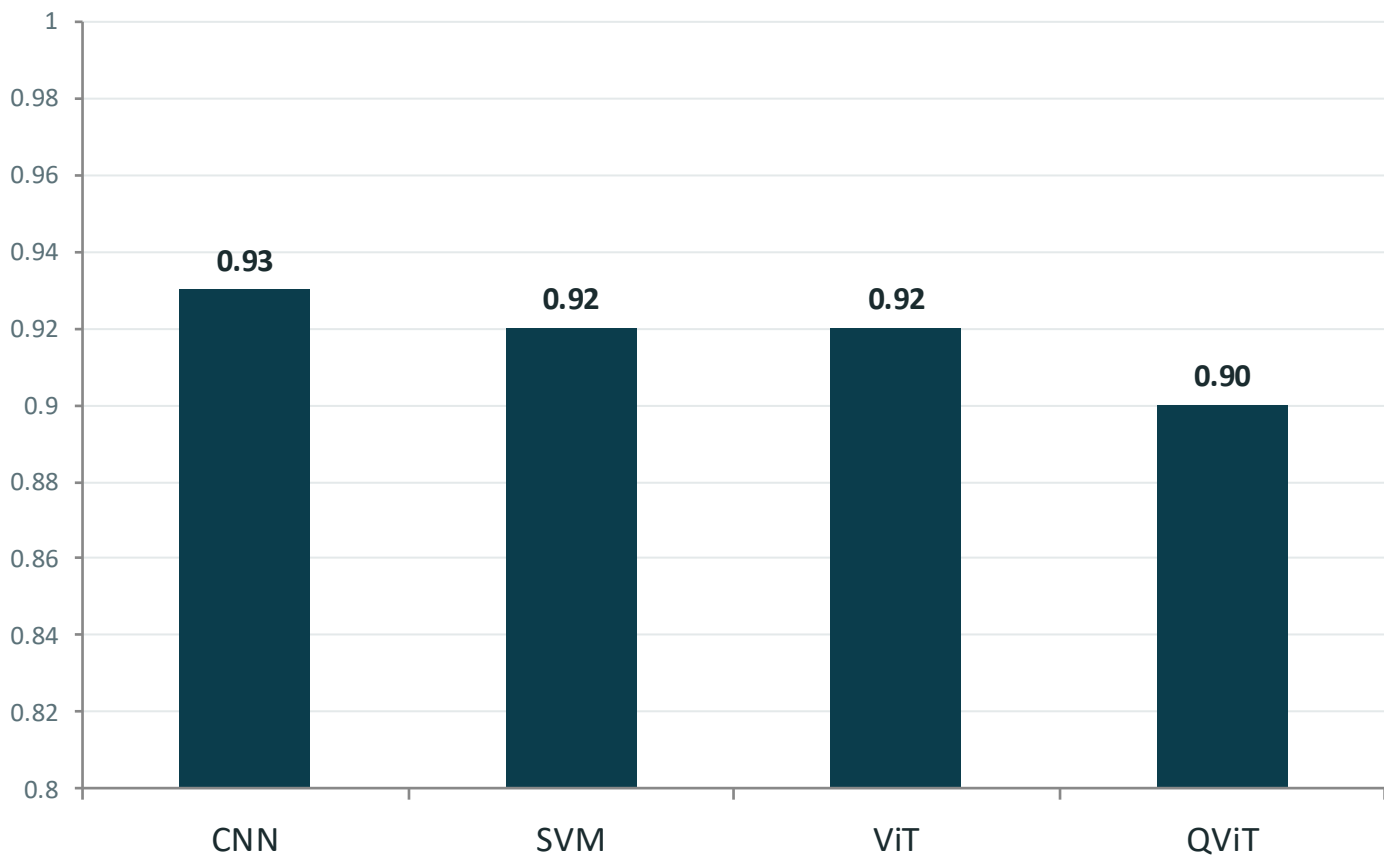
### Design Rationale

QViT was designed to combine ViT's powerful global feature extraction with a variational quantum circuit to explore quantum-enhanced image classification.

A  $768 \rightarrow 64 \rightarrow 12$  bottleneck with GELU compresses ViT embeddings into a quantum-compatible representation. The resulting 12-qubit circuit uses 2 StronglyEntanglingLayers (72 trainable quantum parameters), balancing expressivity against barren-plateau risk on the available data.

## RESULTS

# Cross-Validation Stability (4-Fold, n=252)



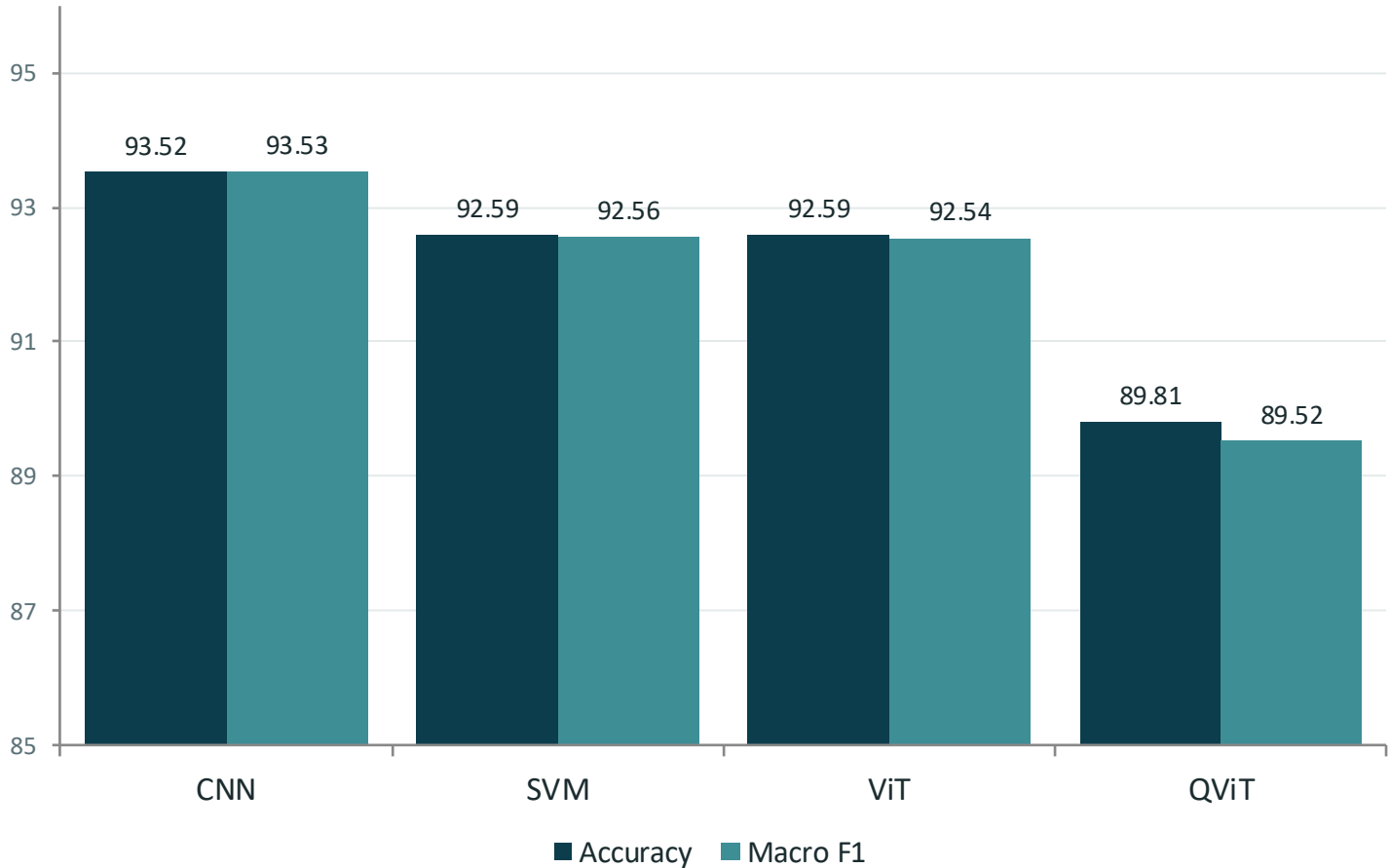
| Model | Mean Acc. ± SD | Mean F1 ± SD | Mean AUC ± SD |
|-------|----------------|--------------|---------------|
| CNN   | 0.92 ± 0.06    | 0.93 ± 0.05  | 0.97 ± 0.02   |
| SVM   | 0.92 ± 0.01    | 0.92 ± 0.01  | 0.96 ± 0.01   |
| ViT   | 0.92 ± 0.01    | 0.92 ± 0.01  | 0.96 ± 0.01   |
| QViT  | 0.90 ± 0.02    | 0.90 ± 0.02  | 0.96 ± 0.03   |

### What drives the variance

- CNN learns from scratch → most sensitive to training-fold composition
- SVM & ViT lean on pretrained representations → most stable
- QViT's 72 quantum parameters are moderately sensitive — more stable than CNN, less than SVM/ViT

RESULTS

# Aggregated Test-Set Performance (n=108)



## CNN Leads

Highest score on all 7 reported metrics — accuracy, precision, recall, F1 (macro & weighted)

## SVM $\approx$ ViT

Nearly identical results (~92.59% acc.) via entirely different mechanisms

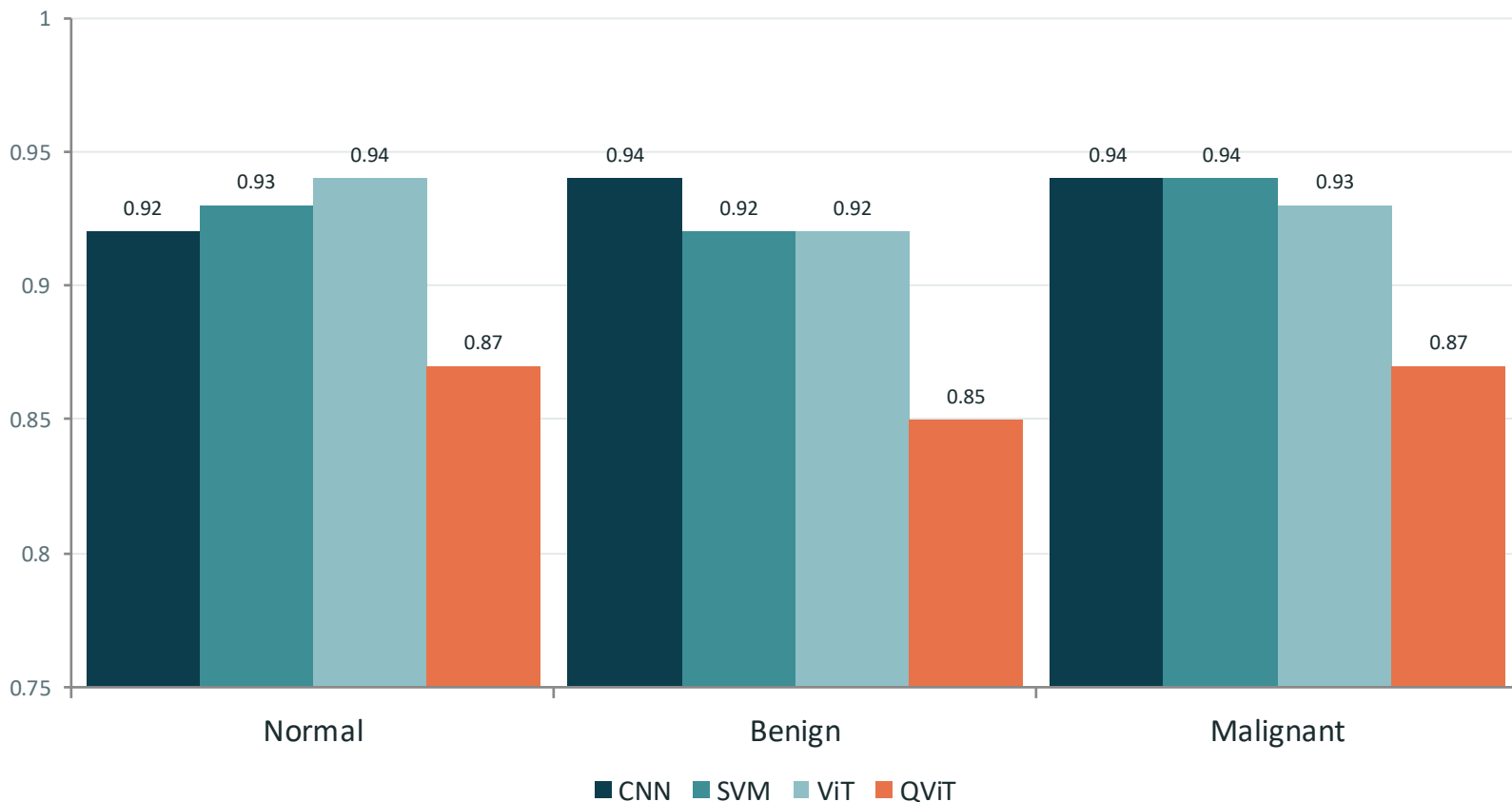
## QViT: -3.71 pts

Trails the CNN by 3.71 accuracy points — a NISQ-era hardware constraint, not a paradigm failure

## RESULTS

# Per-Class Recall — The Clinical Safety Signal

*Recall by diagnostic class (n=36/class) — false negatives carry the highest clinical cost, especially for Malignant cases.*



- **CNN — most balanced:** highest Benign recall (0.94), no weak class
- **ViT — best on Normal:** 0.94 recall, global attention reads healthy tissue well
- **SVM — steady mid-pack:** 0.92–0.94 across all three classes, no standout weakness
- **QViT — safety floor holds, but softest:** QViT misses 15% of Benign cases — flags for radiologist review

# ROC – AUC Analysis — Probabilistic Discrimination

**>0.94**

AUC for every class, across all four models — robust discrimination throughout

**1.00**

Malignant AUC achieved by SVM, ViT, AND QVIT alike

**0.99**

CNN's Benign AUC — the most discriminative Benign probability distribution of any model

**0.97**

CNN's Malignant AUC — slightly below the perfect scores of SVM/ViT/QVIT

**QVIT — ranks correctly, just mis-calibrated: Malignant AUC 1.00 despite 0.87 recall — a threshold fix, not a redesign**

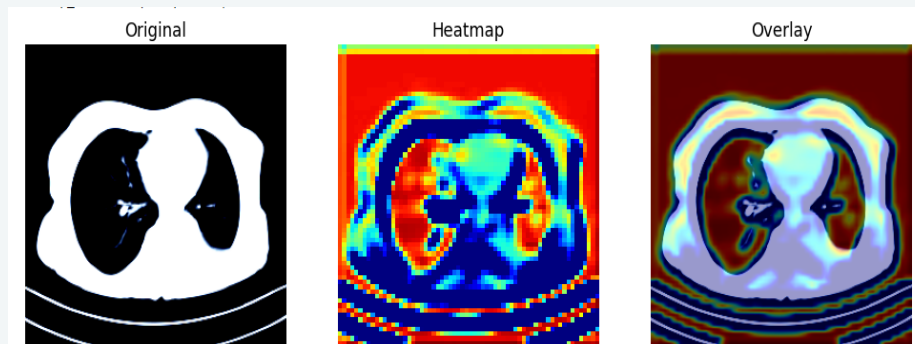
**CNN — confident, not most discriminative: higher hard-label F1 (0.96) but lower AUC (0.97) than SVM**

**SVM — most consistent: perfect Malignant AUC (1.00) sustained across every threshold**

# Interpretability: Do the Models Look in the Right Place?

## CNN

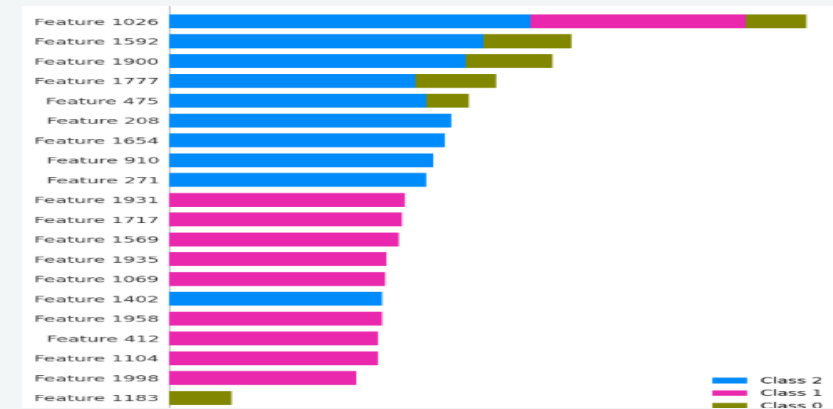
### Grad-CAM Gradient-weighted Class Activation Mapping



Activation concentrates on lung parenchyma and nodule-adjacent tissue;  
minimal response on scanner artefacts

## SVM

### SHAP Shapley Additive Explanations



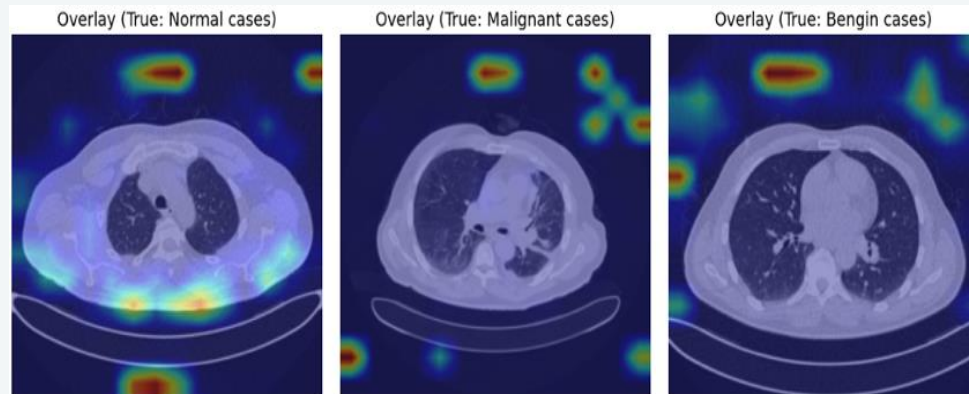
The SHAP analysis confirms that the SVM makes predictions using **meaningful deep feature representations**, rather than acting as a "black-box" classifier.

**Two different forms of transparency:** Grad-CAM gives pixel-level spatial localisation; SHAP gives quantitative feature attribution  
*Both confirm that decisions are grounded in genuine lung pathology rather than scanner artefacts or dataset*

# Interpretability: Do the Models Look in the Right Place?

ViT

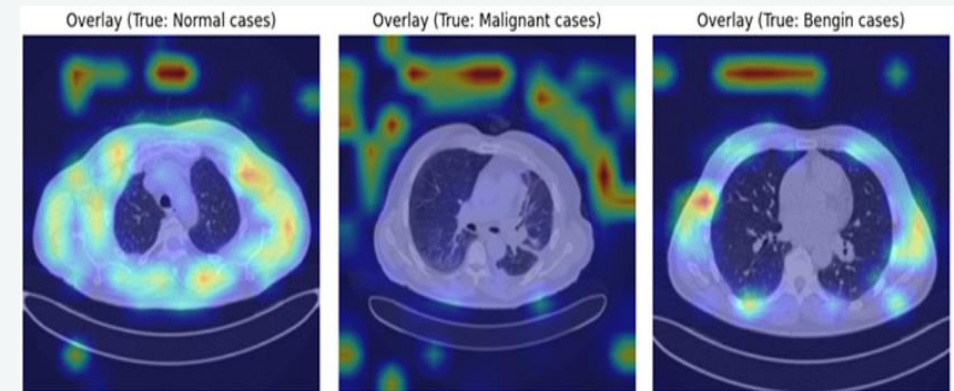
Attention



Spatially coherent focus matching radiologist reasoning — nodule boundaries, bilateral fields for Normal cases contributes to **high classification performance (92.59%)**.

QViT

Attention



Shows **more diffuse and fragmented attention** contributing to **lower Benign (0.85) and Malignant (0.87) recall**

**Directness & clinical verifiability ranking:** CNN (Grad-CAM) ≥ ViT (Attention) > QViT (Attention) > SVM (SHAP)

*Interpretability quality tracks classification performance: the CNN, the top scorer, also produces the most anatomically specific, radiologist-verifiable explanations — model correctness alone is not sufficient for clinical deployment without matching explainability.*

# Comparison with Existing Literature

| Study               | Approach                                         | Accuracy     | Note                                                                             |
|---------------------|--------------------------------------------------|--------------|----------------------------------------------------------------------------------|
| Kareem et al. [7]   | SVM + handcrafted GLCM/Gabor, no balancing       | 89.9%        | This study's SVM: 92.59% (+2.7 pts) via deep transfer features                   |
| Musthafa et al. [9] | CNN + SMOTE oversampling                         | 99.64%       | SMOTE creates synthetic pixel-space blends — not directly comparable             |
| Abe & Nyathi [10]   | CNN Vs ViT + Random Pixel Swap aug.              | 97.56%       | Results depend on augmentation; reported ViT was not the top-performing version. |
| This study          | CNN, SVM, ViT, QViT — unified, balanced protocol | 93.52% (CNN) | First controlled 4-paradigm benchmark on this dataset                            |

*The real contribution isn't a single accuracy number — it's the controlled comparison itself.*  
*Prior single-architecture studies use distinct splits, balancing, and augmentation, making cross-study differences inseparable from experimental design.*  
*Holding every variable constant here isolates architecture as the true source of the observed CNN > SVM ≈ ViT > QViT ordering.*



# Strengths & Limitations

## STRENGTHS

01

### First Controlled Comparison

The first study to benchmark CNN, SVM, ViT, and hybrid QViT on IQ-OTH/NCCD under identical experimental conditions

02

### Interpretable by Design

Grad-CAM, SHAP, and attention maps confirm all four models base decisions on clinically relevant image features

## LIMITATIONS

01

### Single-Institution Dataset

360 balanced images from one hospital and scanner model (Siemens SOMATOM)  
— limits generalisation claims to that acquisition protocol

02

### Simulator, Not Physical Hardware

QViT was evaluated on PennyLane's noise-free analytic simulator, which does not capture decoherence or gate error present on real devices

# Future Research & Practical Implications

## FUTURE RESEARCH

01

### Multi-Centre Validation

External validation on datasets such as LUNA16 and LIDC-IDRI to test generalisation across institutions and scanner manufacturers

02

### Physical NISQ Hardware

Run QVIT on real quantum hardware (IBM Quantum, IonQ) to benchmark accuracy and training stability against the simulator results

## PRACTICAL IMPLICATIONS

01

### Decision-Support Tools

CNN and SVM show strong potential as radiologist decision-support tools, combining high accuracy with practical deployability

02

### Trustworthy Clinical AI

Pairing high predictive performance with explainable AI (Grad-CAM, SHAP, attention) supports development of trustworthy clinical systems

# Key Takeaways

- **CNN** achieved the highest overall performance, demonstrating that convolutional architectures remain highly effective for lung cancer classification on small balanced CT datasets.
- **SVM and Vision Transformer (ViT)** achieved comparable performance, showing that different learning paradigms can produce similar diagnostic accuracy under identical experimental conditions.
- **Hybrid QViT** demonstrated the feasibility of quantum-enhanced learning but remained limited by current NISQ-era hardware constraints.
- **Interpretability analysis** confirmed that all models focused on clinically relevant image features, increasing confidence in their predictions.

*This work establishes the first reproducible, controlled four-paradigm benchmark on the IQ-OTH/NCCD dataset — a template for future comparative studies in quantum medical imaging.*



# Thank You

*Questions & Discussion*

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